

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fifth Semester of B. Tech (CE) Examination****March-April 2018****CE315 Design and Analysis of Algorithm****Date: 29.03.2018, Thursday****Time: 1.30 p.m. To 4.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the questions below.**

- (i) Derive the running time of linear search in worst case. **[03]**
- (ii) Why analysis of algorithm is important? Define: Average case, Best case and Worst case complexity. **[04]**

Q - 2 (a) Attempt the following: (Any Two) **[10]**

- (i) State the Master's theorem with all three cases.
- (ii) Write recurrence relation for Merge sort and solve it using recurrence tree method.
- (iii) Solve the recurrence relation using iteration method: $2T(n-2)+1$
- (b)** Write and Explain Binary Search algorithm for searching value 45 from the following list: 23,36,45,48,52,58,78,90,95 **[04]**
- (c)** Find the optimal solution for the following knapsack problem using fractional approach. Capacity of knapsack is 5. **[04]**

Item No.	Weights	Values
1	1	6
2	2	10
3	3	12

OR

- (c)** Schedule the following jobs to get maximum profit using greedy technique. **[04]**

Job No.	Deadline	Profit
1	3	9
2	2	7
3	3	7
4	1	2

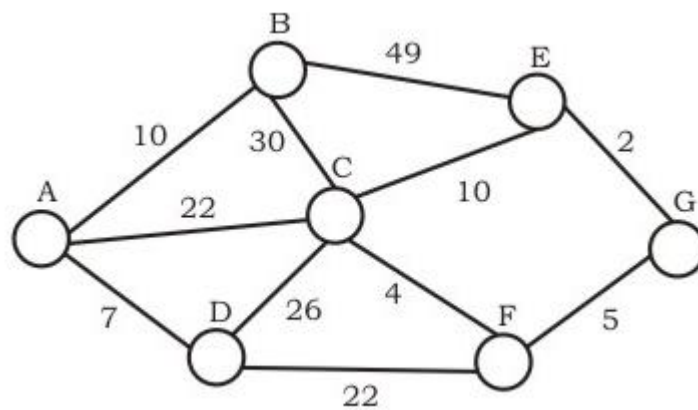
Q – 3 Answer the questions below.

- (i) Sort the sequence 8, 1, 4, 1, 5, 9, 2, 6, 5 using Quick sort method. [05]
- (ii) Does Kruskal's algorithm always give optimal solution? Justify with an example. [05]

OR

Q – 3 Answer the questions below.

- (i) Write and analyze recursive algorithm for finding factorial of a number. [05]
- (ii) Find the minimum spanning tree for the given graph using prim's algorithm. Consider A as starting vertex. [05]



SECTION – II

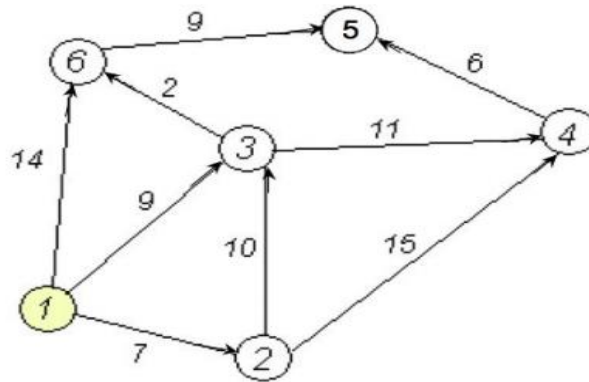
Q - 4 Answer the questions below.

- (i) Explain BFS with example. [03]
- (ii) Compare Backtracking and Branch and Bound technique. [04]

Q - 5 (a) Solve the Assignment Problem with the following cost matrix using Branch & Bound Technique. [05]

	T1	T2	T3
P1	2	4	5
P2	2	7	10
P3	5	3	7

- Q - 5 (b)** Write and Explain Dijkstra's Algorithm for the following graph. Consider 1 as the source vertex. [05]



- Q - 5 (c)** Define the following terms: [04]
- (i) Cross edge
 - (ii) Cycle
 - (iii) Back edge
 - (iv) Tree edge

OR

- Q – 5(a)** Define String Matching Problem. Explain any one String matching algorithm. [05]
- Q – 5 (b)** Find the binomial coefficient $C(10,8)$ using dynamic programming. [05]
- Q – 5 (c)** Explain exponential algorithm for finding 7^{23} . [04]

Q – 6 Attempt the following: (Any Two) [14]

- (i) Consider that the capacity of the knapsack $W = 10$ and the item list is (i_1, i_2, i_3, i_4) . Weights of the items are $(5, 4, 6, 3)$ and values are $(10, 40, 30, 50)$. Solve the 0-1 knapsack problem using Dynamic programming.
- (ii) Find the optimal cost using matrix chain multiplication method for the given matrices: $A=1*2$, $B=2*3$, $C=3*4$, $D=4*3$
- (iii) Compare Dynamic and Greedy methods for making change problem for following example: Total amount is 15 and available coins are of 1, 7, 8, 9
